Abhilash Neog

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Research Interests

Multi-modal learning | Diffusion Models | Scientific Foundation Models | Time-Series Modeling

My research interests lie in cross-modal alignment and multimodal foundation models – spanning LLMs, Vision-Language Models (evaluation, grounding, scientific applications), generative modeling (flow/diffusion models, image editing / generation), and scientific foundation models for irregular and sparse time-series (scaling, reasoning)

Education

Virginia Tech — **Aug 2022 - May 2027**
Ph.D., Computer Science. Advisor: *Dr. Anuj Karpatne*. GPA: 4.0/4.0 — Blacksburg, USA

Virginia Tech — **Aug 2022 - Dec 2024**
M.S., Computer Science. Advisor: *Dr. Anuj Karpatne*. GPA: 4.0/4.0 — Blacksburg, USA

Birla Institute of Technology and Science (BITS), Pilani — **July 2016 – July 2020**
Bachelor of Engineering (B.E.), Computer Science, GPA: 8.08/10 — Pilani, India

Work Experience

Applied Scientist Intern | Microsoft AI — **May 2026 – Aug 2026**

- Proposed a novel training framework for flow-based generative models (FLUX) – via adaptive multiscale wavelet supervision, trained via LoRA fine-tuning – for preserving subject clarity in image outpainting. *Resulted in a research paper (under review)*
- Developed a novel image outpainting pipeline via (a) VLM-based semantic guidance, (b) a high-frequency patch estimation f/w, & (c) an image injection technique for additional conditioning in FLUX models. *Resulted in a patent submission (under review)*

Machine Learning Intern | Kryptowire — **May 2023 – Aug 2023**

- Developed an outlier detection model for denoising sensor-based Human Activity Recognition (HAR) **time series** data
- Built & deployed a CNN-based HAR model achieving 82% F-1 score on an android app using Keras & TensorFlow Lite

Data Scientist | Oracle — **Sep 2020 – July 2022**

- Built & deployed Machine Learning applications into ETL pipelines, leveraging Spark systems, MLOps & CI/CD pipelines
- Designed and deployed a *Demand Prediction* application for time series forecasting using the DeepAR model
- Developed an unsupervised classification algorithm achieving 40% higher accuracy than then LMs on a 71k-label dataset.

Software Engineer Intern | VMware — **Jan 2020 – June 2020**

- Streamlined the process of fetching & filtering raw data from Workspace ONE Cloud using Spring Boot REST APIs
- Contributed to an end-user federation app on Workspace ONE Cloud, and wrote unit tests using JUnit and Mockito

Software Engineer Intern | Samsung Research Institute — **May 2019 – July 2019**

- Performed a feasibility study of Multi-frame Noise Reduction solutions' deployment in Live Focus for Low light conditions
- Optimized the existing HAL call flow, in C++, with considerable noise reduction in the first phase of live focus capture

Research Experience

Graduate Research Assistant | Knowledge-guided Machine Learning Lab — **Jan 2023 – Present**

- Working on cross-modal alignment (LLMs and time-series foundation models) for Q/A, and grounded prediction tasks
- Contributed to a novel physics-guided Diffusion Model for solving Partial Differential Eqns., achieving *15-250x* faster inference
- Zero-shot evaluation of Vision-Language Models (VLMs) for *scene graph generation* and *reasoning ability* on VQA tasks involving scientific images
- Proposed a Model-agnostic Adaptation technique for Irregular and Sparse Time-Series (TS) modeling
- Proposed an *(a) effective tokenization strategy* and *(b) Query-based forecasting*, to train a Foundation Model for spatio-temporal (probabilistic) predictions on irregular grids, leveraging multi-gpu training.

Talks/Awards

- Invited talk at the ASLO 2026 Conference - Session on *AI for the Next Frontier in Aquatic Sciences*
- Invited talk at the AGU 2025 Conference - Session on *Advancing Water Science Through AI*
- Talk on Transfer Learning in Lake Ecosystems at the "*NSF Macrosystems Biology Meeting 2024*"
- Lightning Talk at the "*Frontiers in Ecological Forecasting 2023,*" event at Virginia Tech
- NSF NAIRR (National AI Research Resource) Pilot Award, 2024
- "*Star of the Month (Dec 2021)*" within the Oracle Analytics Cloud Organization, Oracle India

Publications

1. **Abhilash Neog**, Taewan Kim, Yi Wu, Xu Chen, Jian Jiao. "Preserving Subject-Clarity in Image Outpainting with Multiscale Wavelet Supervision". Under Review
2. **Abhilash Neog**, Sepideh Fatemi, Medha Sawhney, et al. "LakeFM: Toward a Foundation Model for Aquatic Ecosystems Using Irregular Multivariate Multi-depth Time Series Data". *KDD 2026* [pdf]
3. **Abhilash Neog**, Arka Daw, Sepideh Fatemi, Medha Sawhney, et al. "Investigating a Model-Agnostic and Imputation-Free Approach for Irregularly-Sampled Multivariate Time-Series Modeling". *TMLR, 2026* [pdf]
4. A. Pandey, **Abhilash Neog**, Gautam Jajoo. "On the Internal Semantics of Time-Series Foundation Models". *BERT2S @ NeurIPS 2025* [pdf]
5. M. Sawhney, **Abhilash Neog**, Mridul Khurana, Anuj Karpatne. "PRISMA: Improving the Accuracy–Latency Frontier of Diffusion-based PDE Solvers Using Physics-Informed Spectral Attention". Under review *(ML4PS @ NeurIPS 2025)* [pdf]
6. S. Fatemi, **Abhilash Neog**, Amartya Dutta, M. Sawhney, et al. "Scientific Equation Discovery using Modular Symbolic Regression via Vision-Language Guidance". *CV4Science @ CVPR 2025 (Oral)*
7. A. Dutta, M. Sawhney, K.S. Mehrab, **Abhilash Neog**, Mridul Khurana, et al. "Open World Scene Graph Generation using Vision Language Models". *World Models @ ICML 2025* [pdf]
8. KS Mehrab, M. Maruf, Arka Daw, **Abhilash Neog**, HB Manogaran, et al. "Fish-Vista: A Multi-Purpose Dataset for Understanding & Identification of Traits from Images". *CVPR 2025* [pdf]
9. M. Maruf, Arka Daw, KS Mehrab, HB Manogaran, **Abhilash Neog**, M. Sawhney, et al. "VLM4Bio: A Benchmark Dataset to Evaluate Pretrained Vision-Language Models for Trait Discovery from Biological Images". *NeurIPS 2024* [pdf]
10. Lavika Goel, **Abhilash Neog**, Ashish Aman, and Arshveer Kaur. "Hybrid Nature-Inspired Optimization Techniques in Face Recognition" *Transactions on Computational Science XXXVI, Springer LNCS, 2020.* [pdf]

Patents

1. **Abhilash Neog**, Taewan Kim, Yi Wu, Xu Chen. Microsoft, 2026. "Subject-Clarity-Aware Image Outpainting via Semantic and Structural Guidance with Vision-Language Models and High-Frequency Map Estimation." *Patent submission (under review).*
2. Akash B., Krishnan R., **Abhilash, N.**, Dipawesh P. and Karthik B. Oracle, 2024. "Machine Learning Based Spend Classification." *U.S. Patent Application 17/903,161.* [pdf]

Selected Projects

Can Large Vision Language Models (VLMs) Ground Fine-grained Attribute? **pdf** — **Aug '24 – Dec '24**
- Developed a novel dual-scale attention framework for fine-grained attribute localization in Large Vision-Language Models (LVLMs), incorporating entropy-based head selection, maximally connected component filtering, and hierarchical constraints

Evaluating Model Reasoning & Hallucinations in Medical LLMs **code** **pdf** — **Jan '24 – April '24**
- Analyzed and evaluated factual error propagation in open-source medical LLMs such as BioMistral, Asclepius, Alpacare, and PMC-LLaMA to identify variations in their efficacy and ensure reliable information dissemination in medical settings.

Mathematical Reasoning in Large Language Models (LLMs) **code** **pdf** — **Aug '23 – Dec '23**
- Worked on the problem of numerical headline generation and numeral masked-fill as part of NumEval @ SemEval 2024
- Adapted **Llama**, **T5**, **BART** & **RoBERTa** models by Direct **fine-tuning** & **prompt engineering** for the respective tasks

Text Summarization of Electronic Theses and Dissertations (ETD) **pdf** — **Sept '22 – Dec '22**
- Developed a text summarization pipeline, integrating both Transformer-based abstractive algorithms (pre-trained Pegasus & RoBERTa) and traditional extractive algorithms like TextRank, LexRank & LSA, within an ETD Info. Retrieval system

Academic Duties

- **Reviewing**:
  - Conferences/Journal - NeurIPS ('25, '26), AAAI ('26, '27), TMLR, ICML 2026, KDD 2026, ECCV 2026
  - Workshops - FM4Science (ICLR '26), FMSD (ICML '25, '26), FMs in the Wild (ICLR '25), Open Science for FMs (ICLR '25), XAI4Science (ICLR '25), ML4PS (NeurIPS '25), TSALM (NeurIPS '24)
- **Teaching**:
  - Graduate Teaching Assistant, CS 5805 Machine Learning, Spring 2024
  - Teaching Assistant, BITS F312 Neural Networks and Fuzzy Logic, Fall 2019

Technical Skills

Languages: Python, Java, C++, SQL, R

Frameworks: PyTorch, Git, Hydra, Tensorflow Keras, Spark